

Statistics for Social Science Research

Lecture 4

24 August 2026

Topics for today

- Probability (Chapter 6, Statistics for the Behavioral Sciences)
- Probability and Samples: The Distribution of Sample Means (Chapter 7, Statistics for the Behavioral Sciences)

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

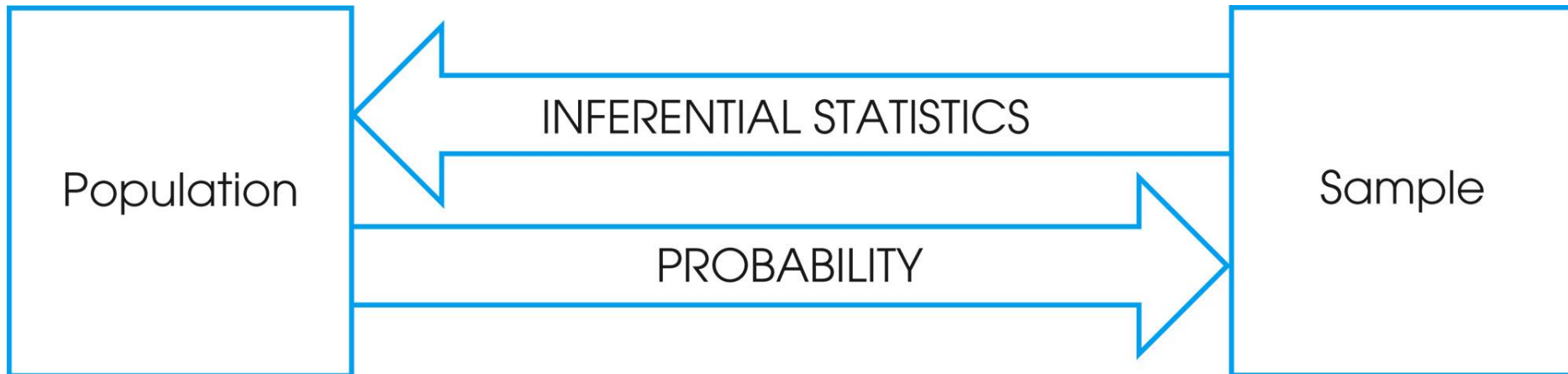
Chapter 6

Probability

6.1 Introduction to Probability

- Research begins with a question about an entire population.
- Actual research is conducted using a sample.
- Inferential statistics use sample data to answer questions about the population
- Relationships between samples and populations are defined in terms of probability

Figure 6.1 Role of probability in inferential statistics



Definition of Probability

- Several different outcomes are possible
- The probability of any specific outcome is a fraction or proportion of all possible outcomes

$$\text{probability of } A = \frac{\text{number of outcomes classified as } A}{\text{total number of possible outcomes}}$$

Probability Notation

- p is the symbol for “probability”
- Probability of some specific outcome is specified by $p(\text{event})$
- So the probability of drawing a red ace from a standard deck of playing cards could be symbolized as $p(\text{red ace})$
- Probabilities are always proportions
- $p(\text{red ace}) = 2/52 \approx 0.03846$ (proportion is 2 red aces out of 52 cards)

(Independent) Random Sampling

- A process or procedure used to draw samples
- Required for our definition of probability to be accurate
- The “Independent” modifier is generally left off, so it becomes “random sampling”

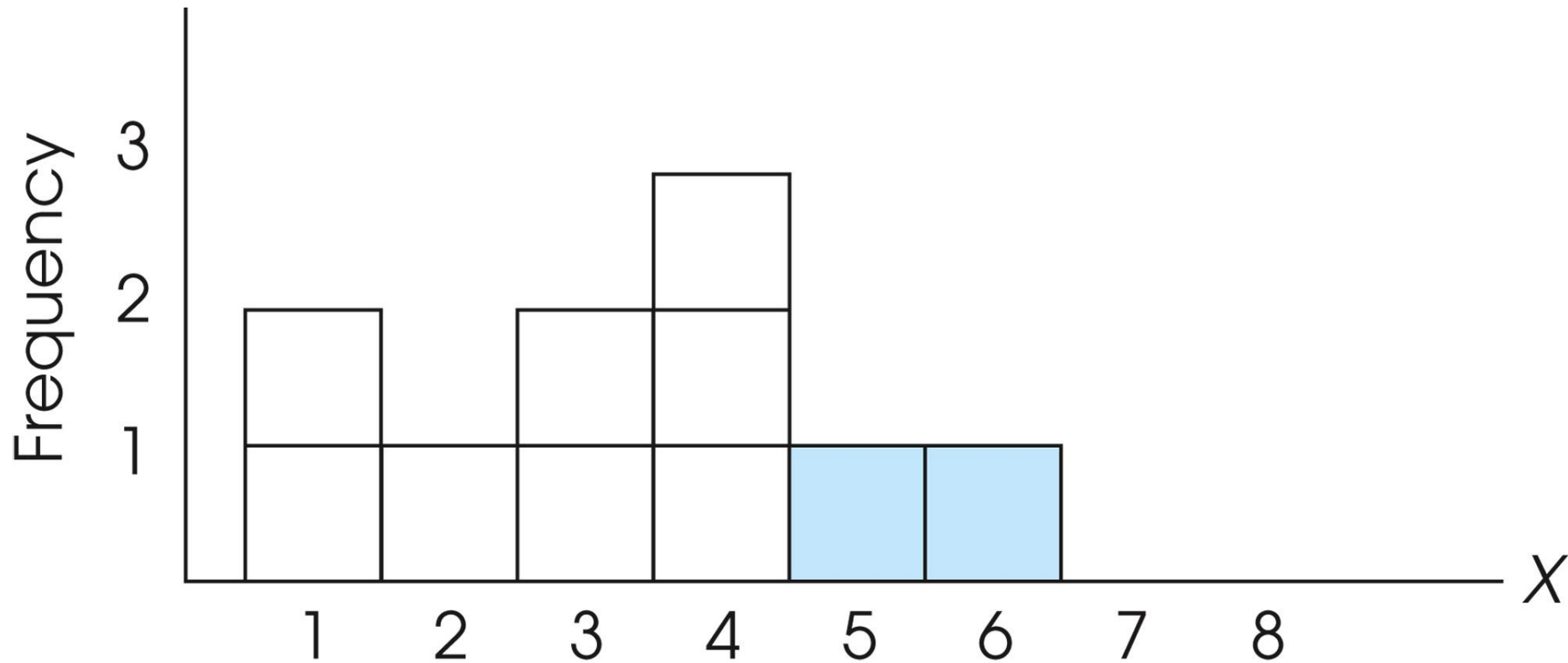
Definition of Random Sample

- A sample produced by a process that assures:
 - Each individual in the population has an equal chance of being selected
 - Probability of being selected stays constant from one selection to the next when more than one individual is selected
- Requires sampling with replacement

Probability and Frequency Distributions

- Probability usually involves population of scores that can be displayed in a frequency distribution graph
- Different portions of the graph represent portions of the population
- Proportions and probabilities are equivalent
- A particular portion of the graph corresponds to a particular probability in the population

Figure 6.2 Population
Frequency Distribution Histogram



Learning Check

- A deck of 52 cards contains 12 royalty cards. If you randomly select a card from the deck, what is the probability of obtaining a royalty card?

A

• $p = 1/52$

B

• $p = 12/52$

C

• $p = 3/52$

D

• $p = 4/52$

Learning Check - Answer

- A deck of 52 cards contains 12 royalty cards. If you randomly select a card from the deck, what is the probability of obtaining a royalty card?

A

• $p = 1/52$

B

• $p = 12/52$

C

• $p = 3/52$

D

• $p = 4/52$

Learning Check TF

- Decide if each of the following statements is **True** or **False**.

T/F

- Choosing random individuals who walk by yields a random sample

T/F

- Probability predicts what kind of population is likely to be obtained

Learning Check - Answers

False

- Not all individuals walk by, so not all have an equal chance of being selected for the sample

False

- The population is given. Probability predicts what a sample is likely to be like

6.2 Probability and the Normal Distribution

- Normal distribution is a common shape
 - Symmetrical
 - Highest frequency in the middle
 - Frequencies taper off towards the extremes
- Defined by an equation
- Can be described by the proportions of area contained in each section.
- z-scores are used to identify sections

Figure 6.3

The Normal Distribution

$$Y = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(X-\mu)^2/2\sigma^2}$$

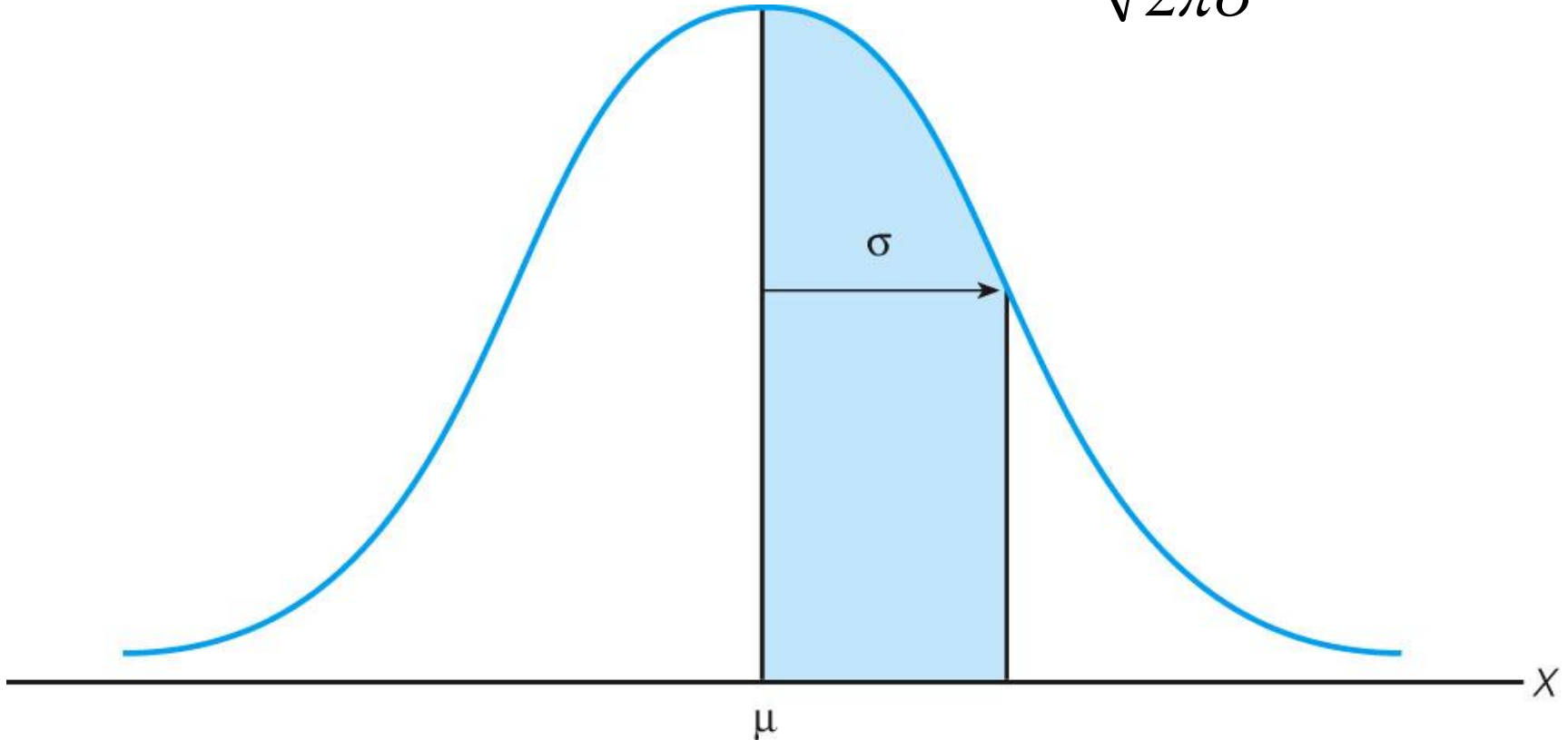
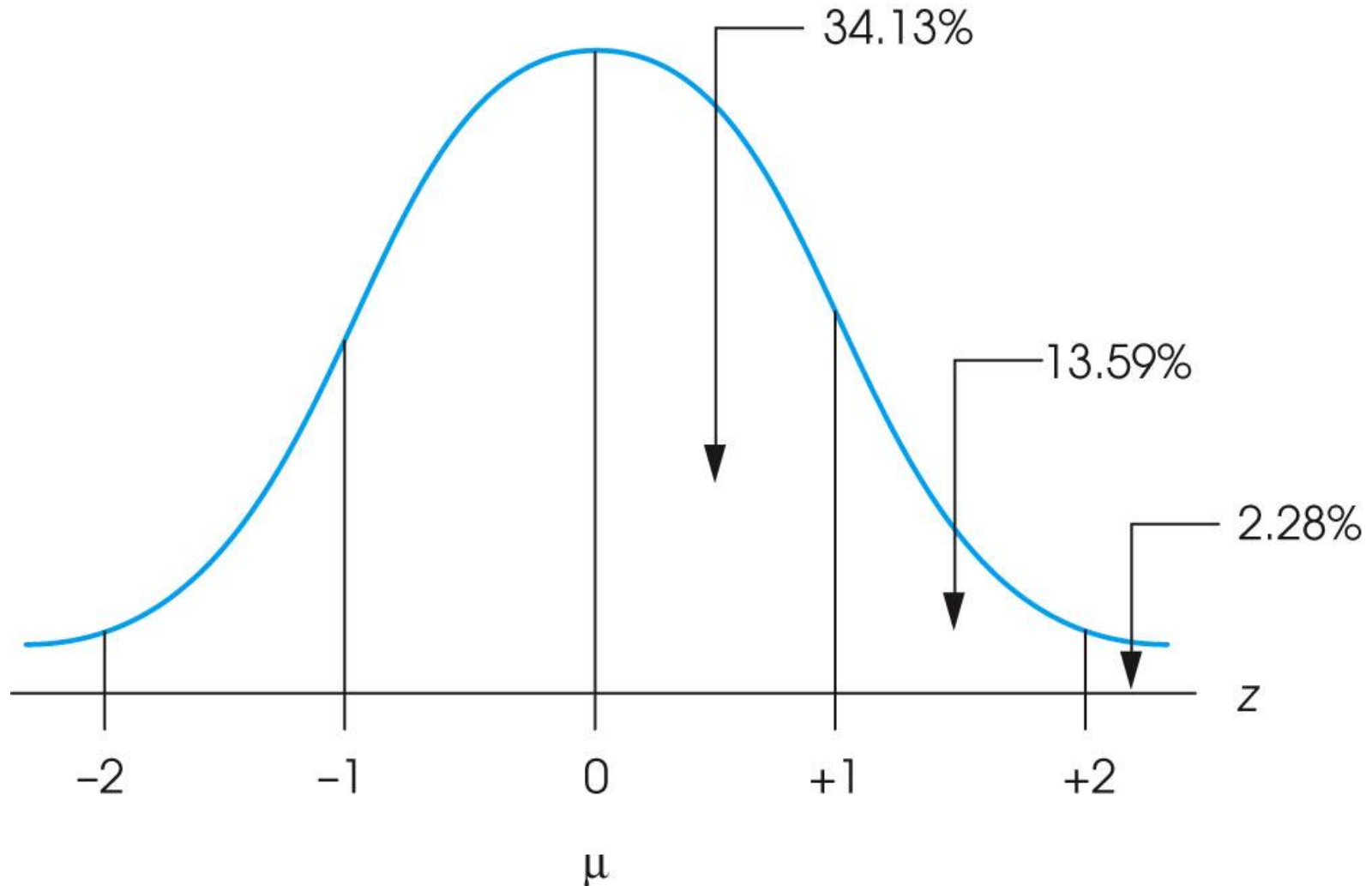


Figure 6.4

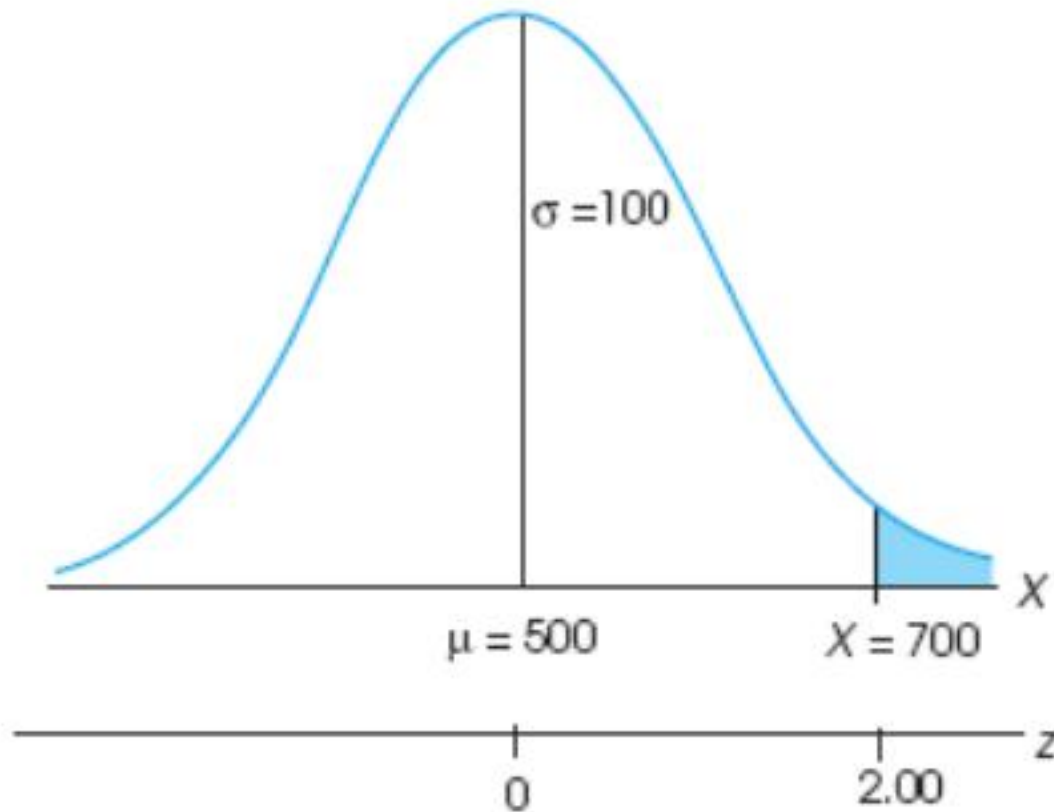
Normal Distribution with z-scores



Characteristics of the Normal Distribution

- Sections on the left side of the distribution have the same area as corresponding sections on the right
- Because z-scores define the sections, the proportions of area apply to *any normal distribution*
 - Regardless of the mean
 - Regardless of the standard deviation

Figure 6.5
Distribution for Example 6.2



The Unit Normal Table

- The proportion for only a few z-scores can be shown graphically
- The complete listing of z-scores and proportions is provided in the *unit normal table*
- Unit Normal Table is provided in Appendix B, Table B.1

Figure 6.6

Portion of the Unit Normal Table

(A) z	(B) Proportion in body	(C) Proportion in tail	(D) Proportion between mean and z
0.00	.5000	.5000	.0000
0.01	.5040	.4960	.0040
0.02	.5080	.4920	.0080
0.03	.5120	.4880	.0120
0.21	.5832	.4168	.0832
0.22	.5871	.4129	.0871
0.23	.5910	.4090	.0910
0.24	.5948	.4052	.0948
0.25	.5987	.4013	.0987
0.26	.6026	.3974	.1026
0.27	.6064	.3936	.1064
0.28	.6103	.3897	.1103
0.29	.6141	.3859	.1141
0.30	.6179	.3821	.1179
0.31	.6217	.3783	.1217
0.32	.6255	.3745	.1255
0.33	.6293	.3707	.1293
0.34	.6331	.3669	.1331

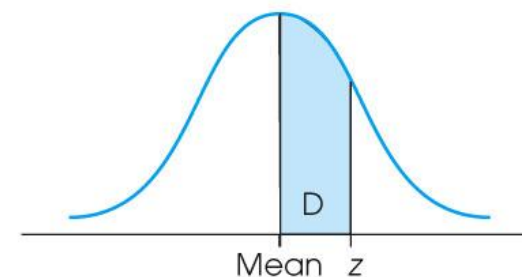
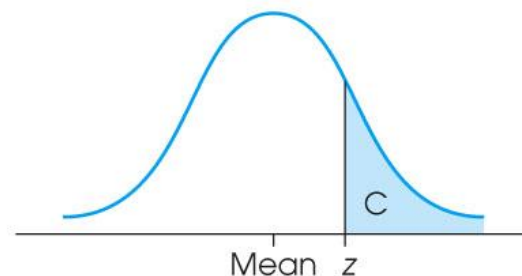
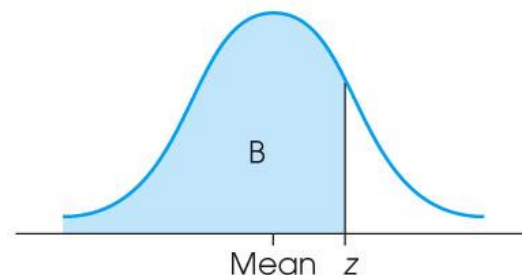
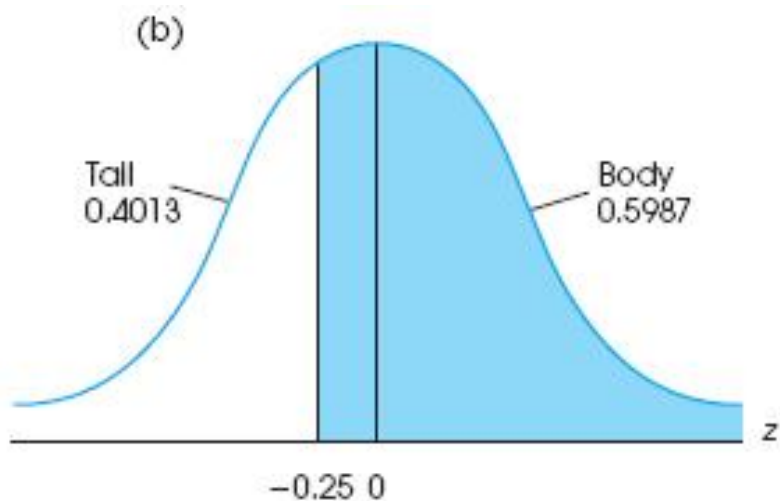
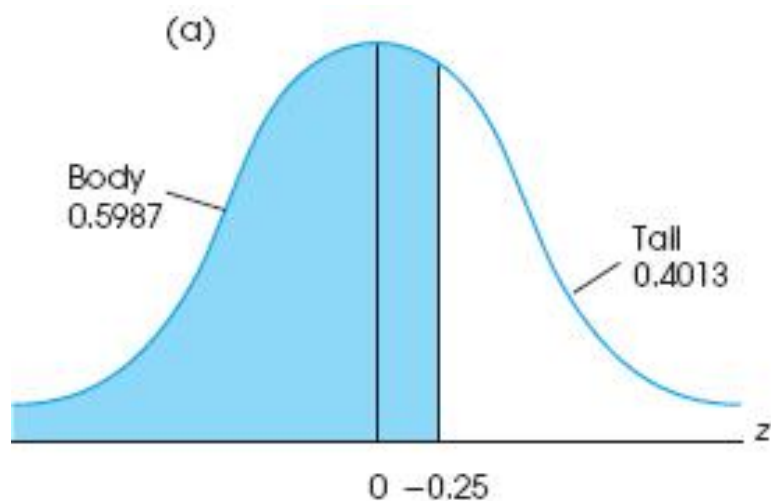


Figure 6.7 Proportions Corresponding to $z = \pm 0.25$



Probability/Proportion & z-scores

- Unit normal table lists relationships between z-score locations and proportions in a normal distribution
- If you know the z-score, you can look up the corresponding proportion
- If you know the proportion, you can use the table to find a specific z-score location
- Probability is equivalent to proportion

Figure 6.8

Distributions: Examples 6.3a—6.3c

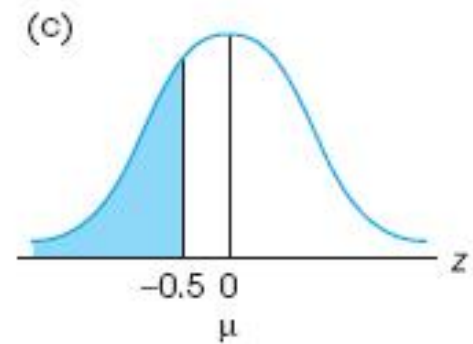
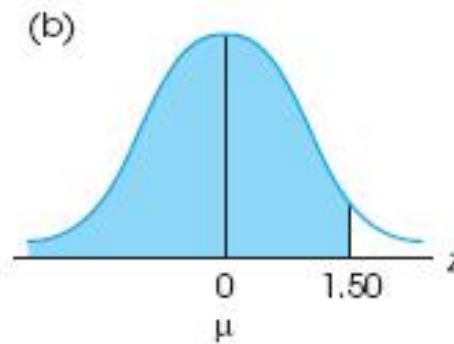
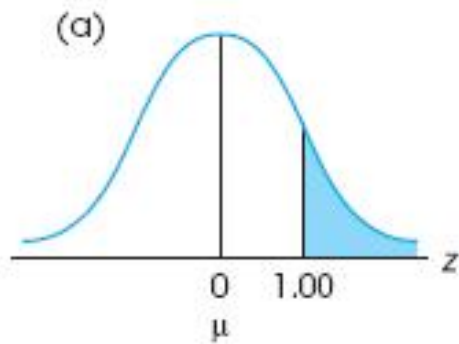
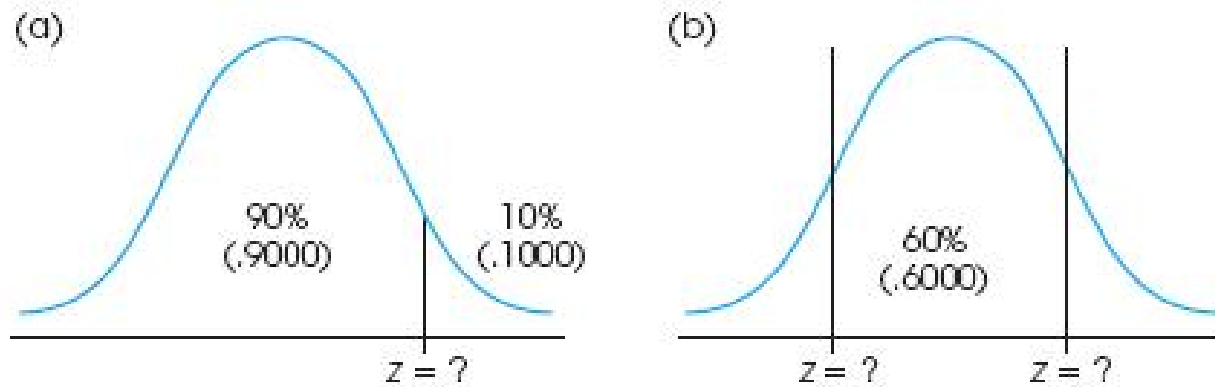


Figure 6.9

Distributions: Examples 6.4a—6.4b



Learning Check

- Find the proportion of the normal curve that corresponds to $z > 1.50$

A

• $p = 0.9332$

B

• $p = 0.5000$

C

• $p = 0.4332$

D

• $p = 0.0668$

Learning Check - Answer

- Find the proportion of the normal curve that corresponds to $z > 1.50$

A

• $p = 0.9332$

B

• $p = 0.5000$

C

• $p = 0.4332$

D

• $p = 0.0668$

Learning Check

- Decide if each of the following statements is **True** or **False**.

T/F

- For any negative z-score, the tail will be on the right hand side

T/F

- If you know the probability, you can find the corresponding z-score

Learning Check - Answer

False

- For negative z-scores the tail will *always* be on the left side

True

- First find the proportion in the appropriate column then read the z-score from the left column

6.3 Probabilities/Proportions for Normally Distributed Scores

- The probabilities given in the Unit Normal Table will be accurate ***only*** for normally distributed scores so the shape of the distribution should be verified before using it.
- For normally distributed scores
 - Transform the X scores (values) into z -scores
 - Look up the proportions corresponding to the z -score values.

Figure 6.10
Distribution of IQ scores

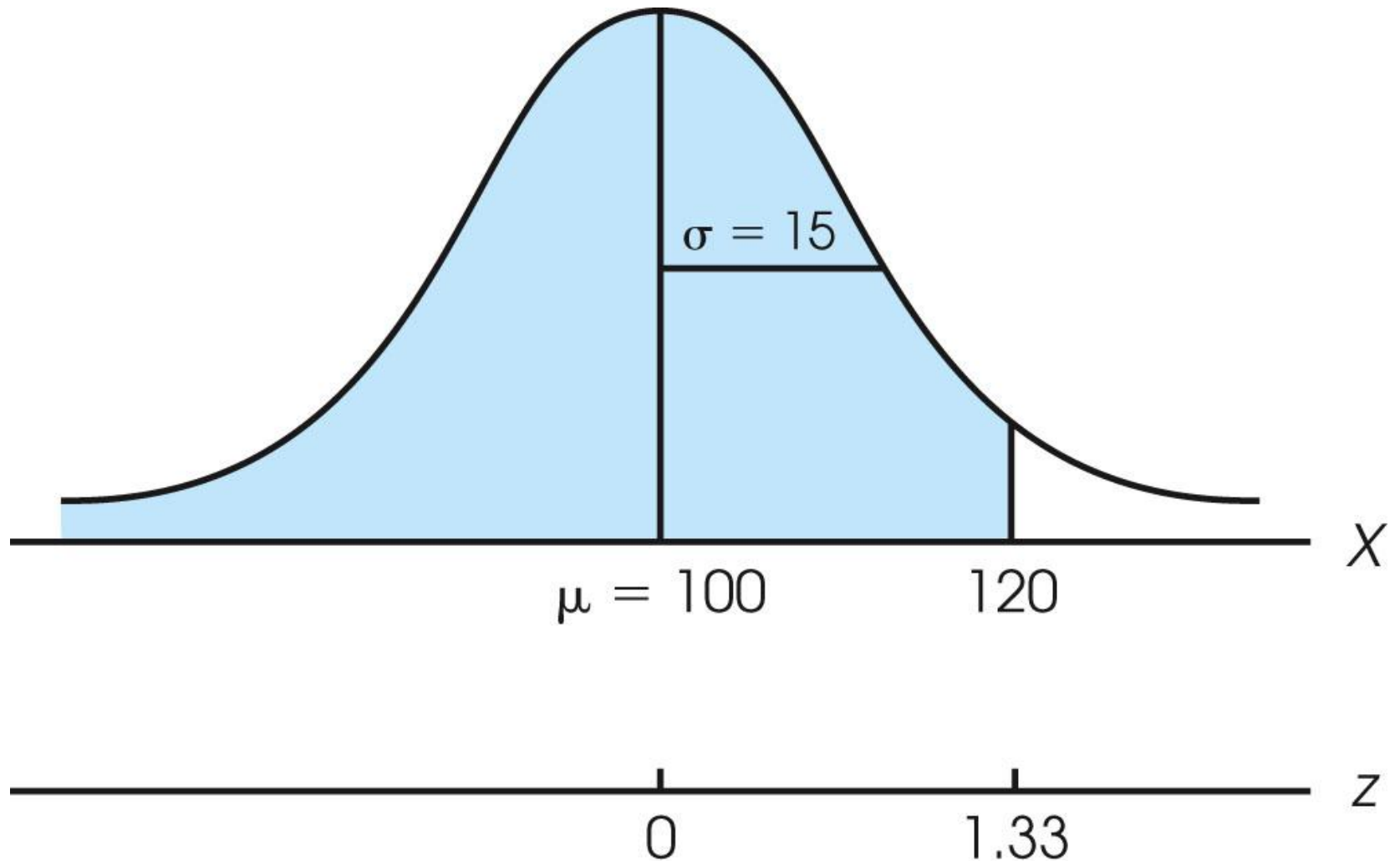
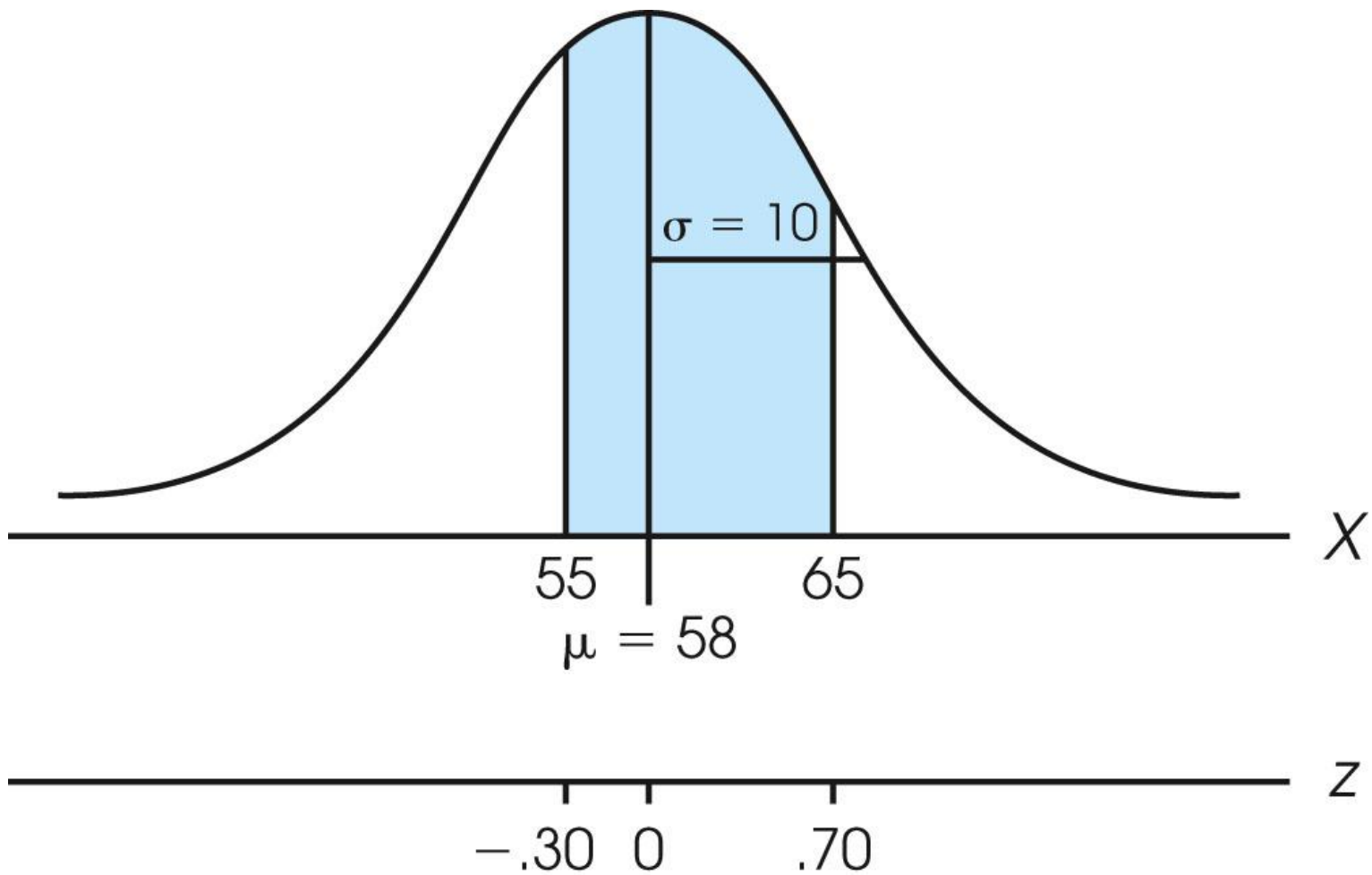


Figure 6.11
Example 6.6 Distribution



Box 6.1 Percentile ranks

- Percentile rank is *the percentage of individuals in the distribution who have scores that are less than or equal to the specific score.*
- Probability questions can be rephrased as percentile rank questions.

Figure 6.12
Example 6.7 Distribution

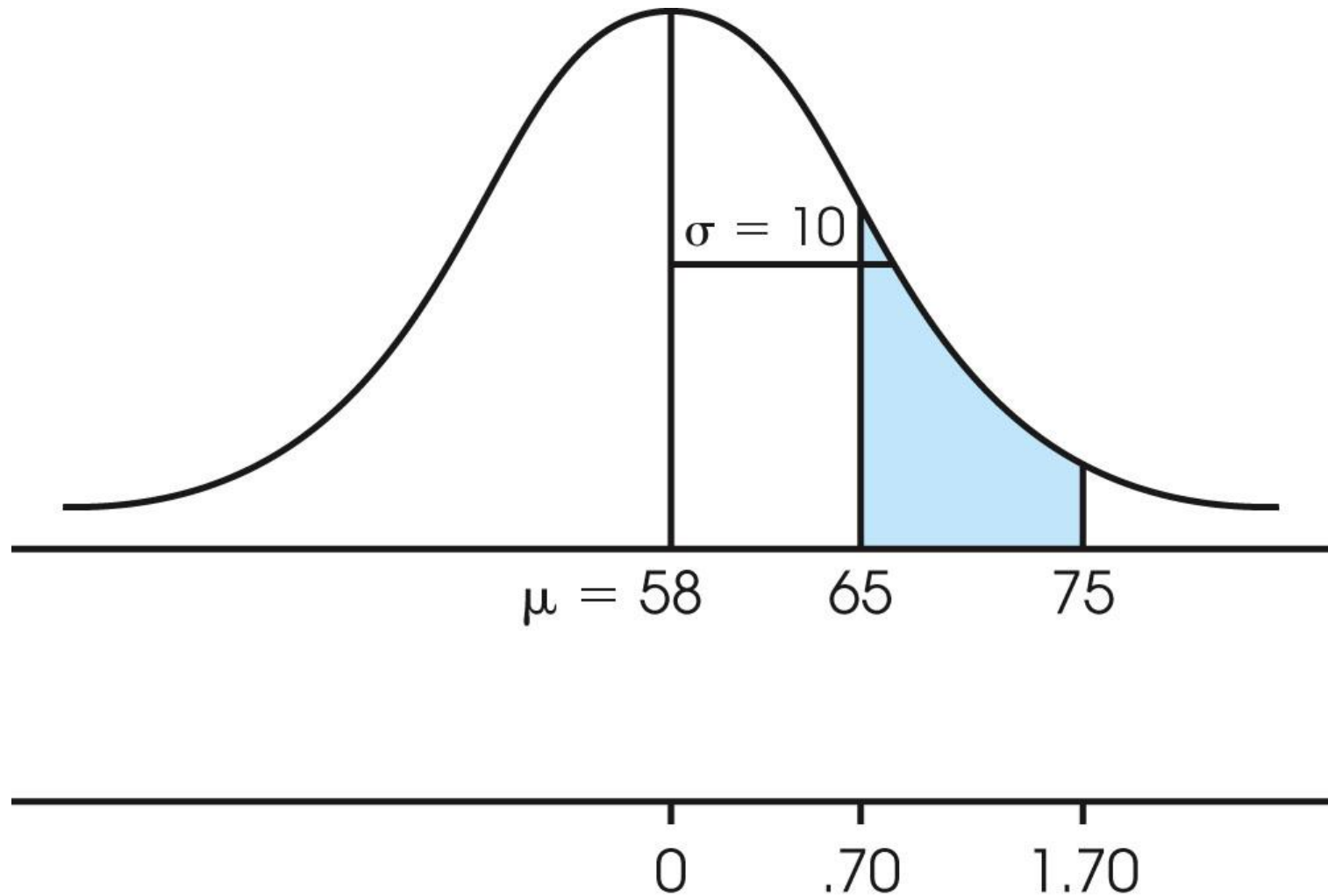


Figure 6.13 Determining Normal Distribution Probabilities/Proportions

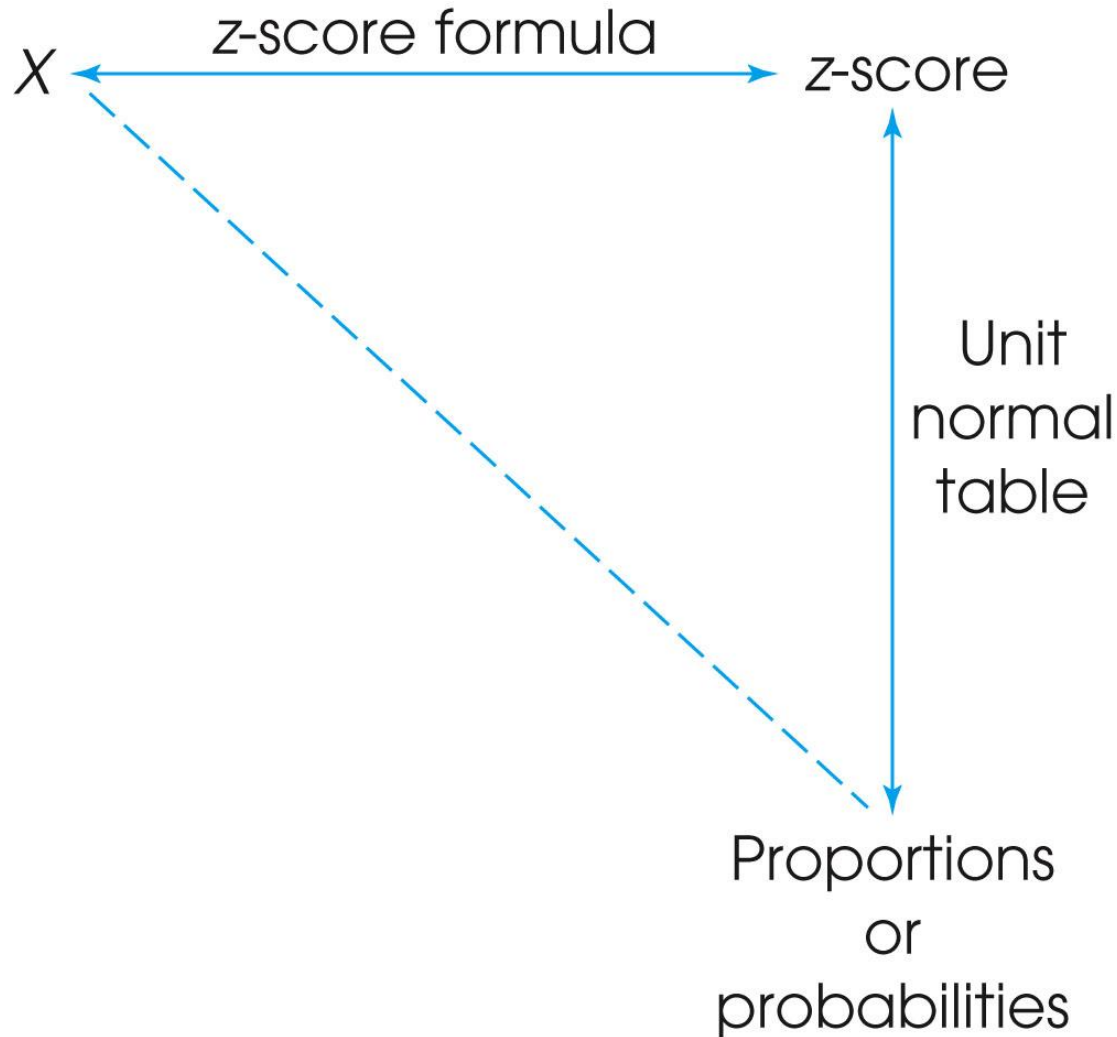


Figure 6.14

Commuting Time Distribution

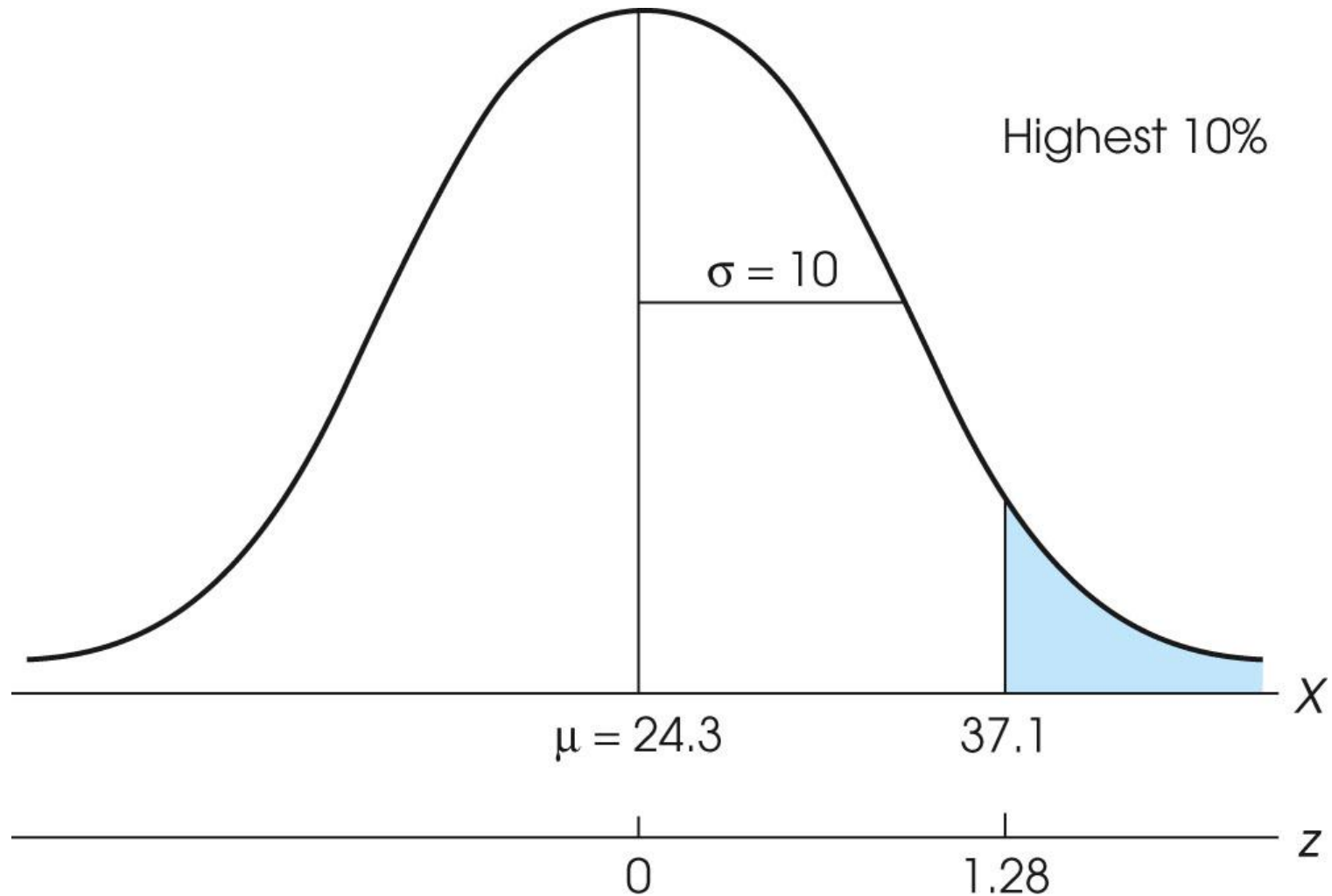
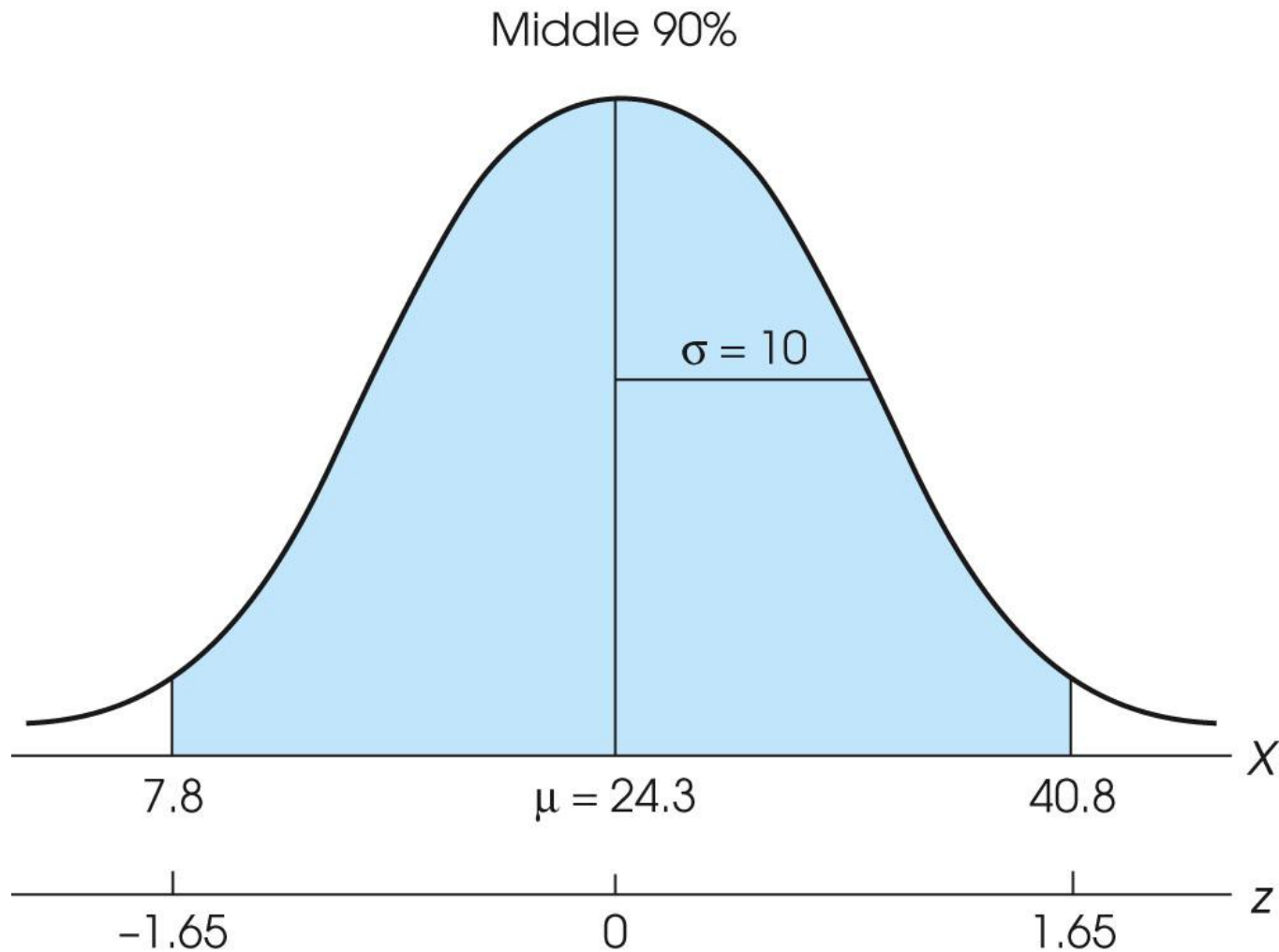


Figure 6.15

Commuting Time Distribution



Learning Check

- Membership in MENSA requires a score of 130 on the Stanford-Binet 5 IQ test, which has $\mu = 100$ and $\sigma = 15$. What proportion of the population qualifies for MENSA?

A

• $p = 0.0228$

B

• $p = 0.9772$

C

• $p = 0.4772$

D

• $p = 0.0456$

Learning Check - Answer

- Membership in MENSA requires a score of 130 on the Stanford-Binet 5 IQ test, which has $\mu = 100$ and $\sigma = 15$. What proportion of the population qualifies for MENSA?

A • $p = 0.0228$

B • $p = 0.9772$

C • $p = 0.4772$

D • $p = 0.0456$

Learning Check

- Decide if each of the following statements is True or False.

T/F

- It is possible to find the X score corresponding to a percentile rank in a normal distribution

T/F

- If you know a z -score you can find the probability of obtaining that z -score in a distribution of any shape

Learning Check - Answer

True

- Find the z-score for the percentile rank, then transform it to X

False

- If a distribution is skewed the probability shown in the unit normal table will *not* be accurate

6.4 Looking Ahead to Inferential Statistics

- Many research situations begin with a population that forms a normal distribution
- A random sample is selected and receives a treatment, to evaluate the treatment
- Probability is used to decide whether the treated sample is “noticeably different” from the population

Figure 6.16

Research Study Conceptualization

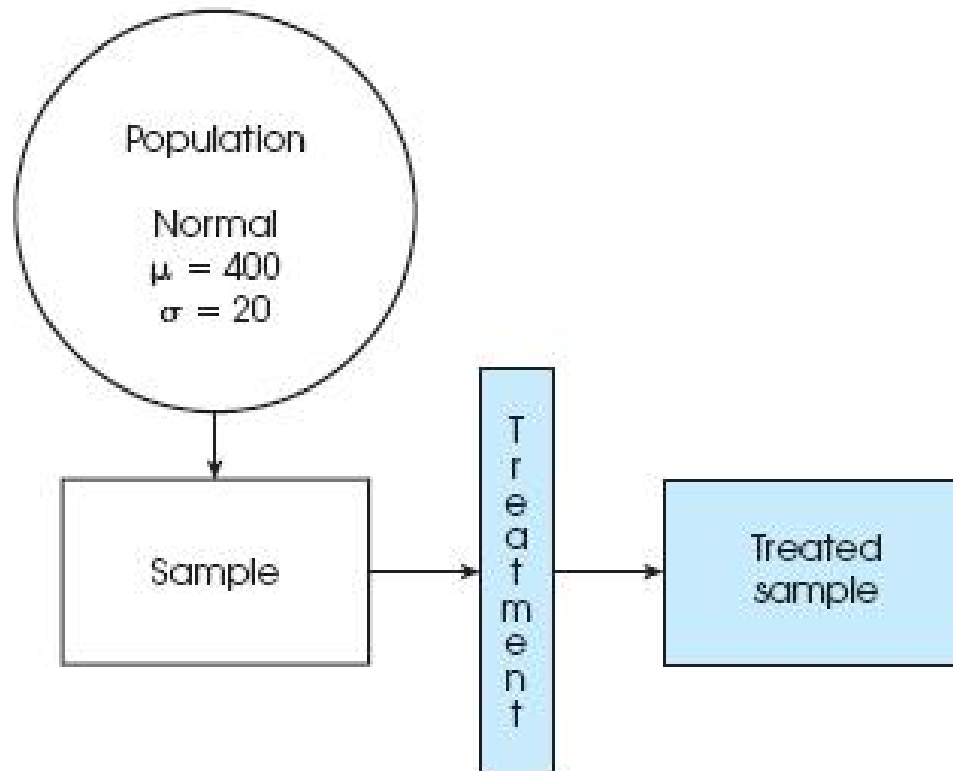


Figure 6.17

Research Study Conceptualization

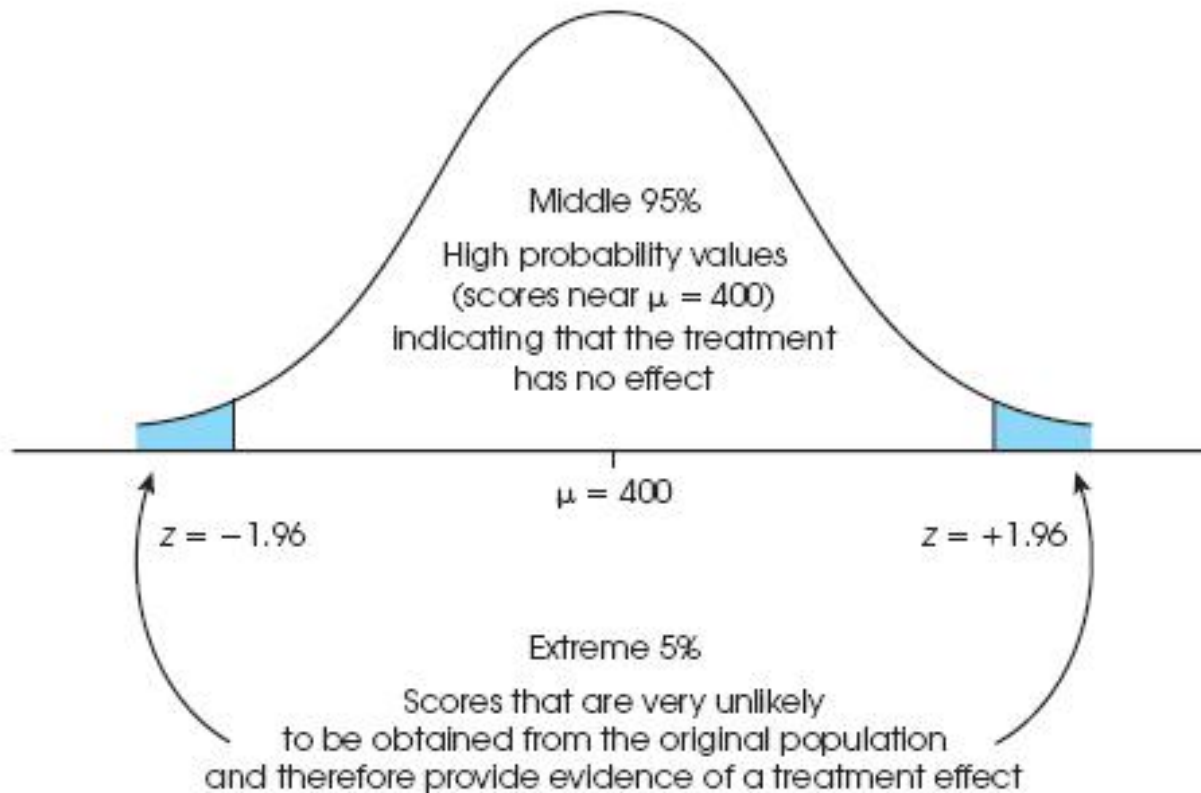
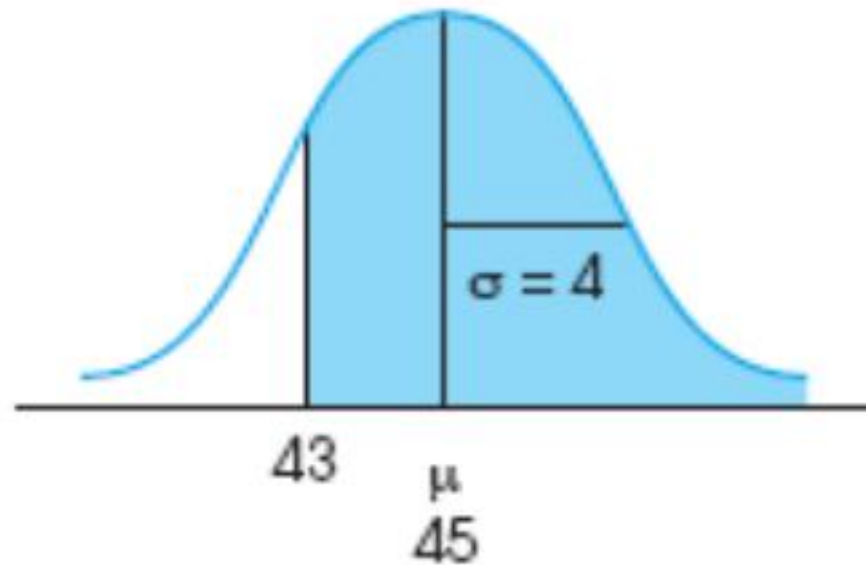


Figure 6.18

Demonstration 6.1



Chapter 7

Probability and Samples: The Distribution of Sample Means

7.1 Samples and Population

- The location of a score in a sample or in a population can be represented with a z-score
- Researchers typically want to study entire samples rather than single scores
- Sample provides estimate of the population
- Tests involve transforming sample mean to a z-score

Sampling Error

- Error does not indicate a mistake was made
- Sampling error is the natural discrepancy, or the amount of error, between a sample statistic and its corresponding population parameter
- Samples are variable; two samples are *very, very rarely identical*

7.2 Distribution of Sample Means

- Samples differ from each other
 - Given a random sample it is unlikely that sample means would always be the same
 - Sample means differ from each other
- The distribution of sample means is the collection of sample means for all the possible random samples of a particular size (n) that can be obtained from a population

Sampling distribution

- Distributions in earlier chapters were distributions of scores from samples
- “Distribution of Sample Means” is called a sampling distribution
 - The “Distribution of Sample Means” is a special kind of population
 - It is a distribution of sample means obtained by selecting all the possible samples of a specific size (n) from a population

Figure 7.1 Population
Frequency Distribution Histogram

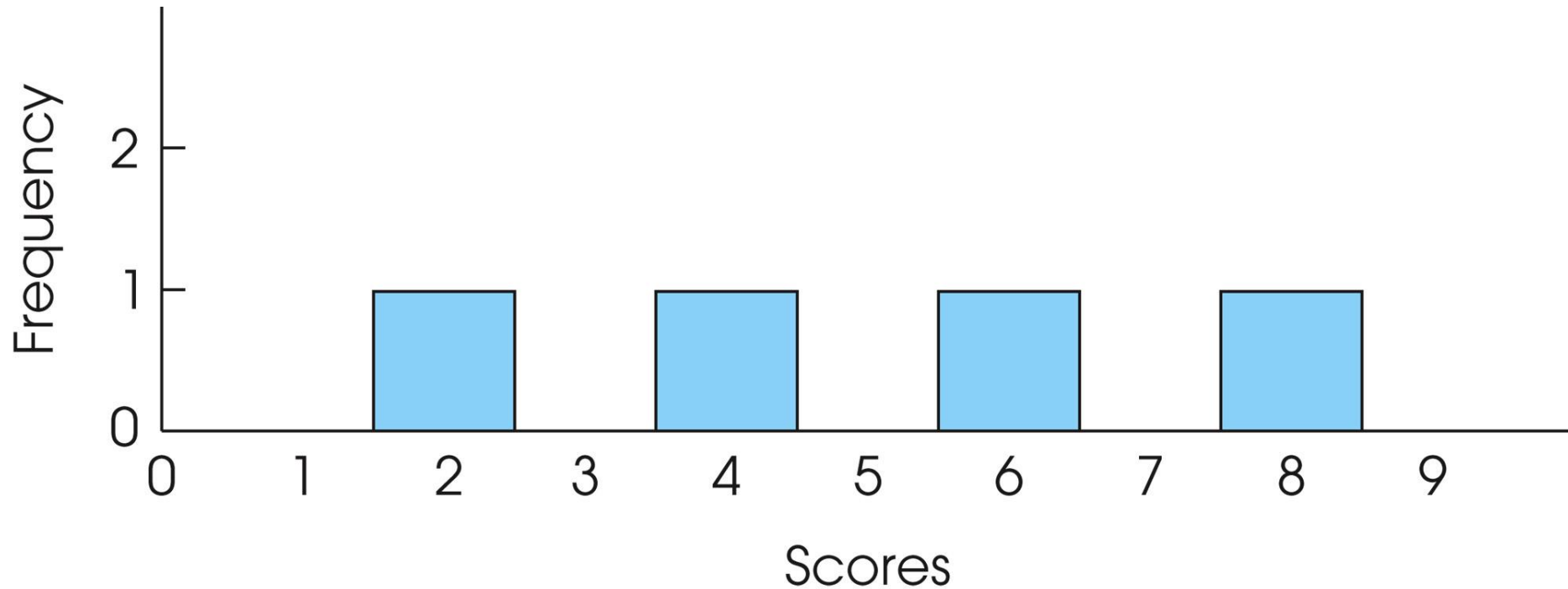
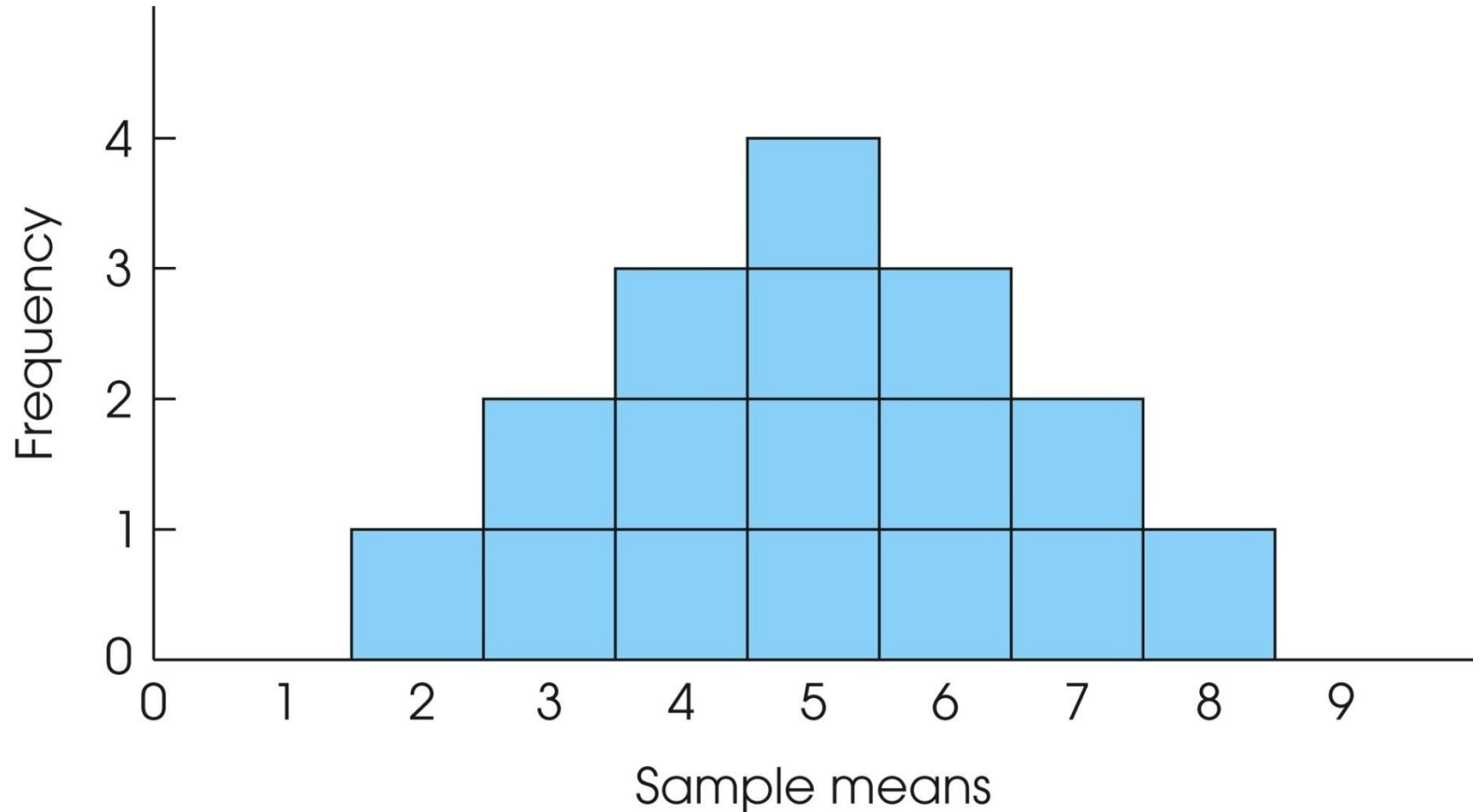


Figure 7.2 Distribution of
Sample Means ($n=2$)



Important Characteristics of Distributions of Sample Means

- The sample means pile up around the population mean
- The distribution of sample means is approximately normal in shape

Central Limit Theorem

- Applies to any population with mean μ and standard deviation σ
- Distribution of sample means approaches a normal distribution as n approaches infinity
- Distribution of sample means for samples of size n will have a mean of μ_M
- Distribution of sample means for samples of size n will have a standard deviation = $\frac{\sigma}{\sqrt{n}}$

Shape of the Distribution of Sample Means

- The distribution of sample means is almost perfectly normal in either of two conditions
 - The population from which the samples are selected is a normal distribution
 - or
 - The number of scores (n) in each sample is relatively large—at least 30

Expected Value of M

- Mean of the distribution of sample means is μ_M and has a value equal to the mean of the population of scores, μ
- Mean of the distribution of sample means is called the ***expected value*** of M
- M is an unbiased statistic because μ_M , the expected value of the distribution of sample means is the value of the population mean, μ

Standard Error of M

- Variability of a distribution of scores is measured by the standard deviation
- Variability of a distribution of sample means is measured by the standard deviation of the sample means, and is called the standard error of M and written as σ_M
- In journal articles or other textbooks, the standard error of M might be identified as “standard error,” “SE,” or “SEM”

Standard Error of M

- The standard error of M is the standard deviation of the distribution of sample means
- The standard error of M provides a measure of how much distance is expected on average between M and μ

Standard Error Magnitude

- Law of large numbers: the larger the sample size, the more probable it is that the sample mean will be close to the population mean
- Population variance: The smaller the variance in the population, the more probable it is that the sample mean will be close to the population mean

Figure 7.3 Standard Error and Sample Size Relationship

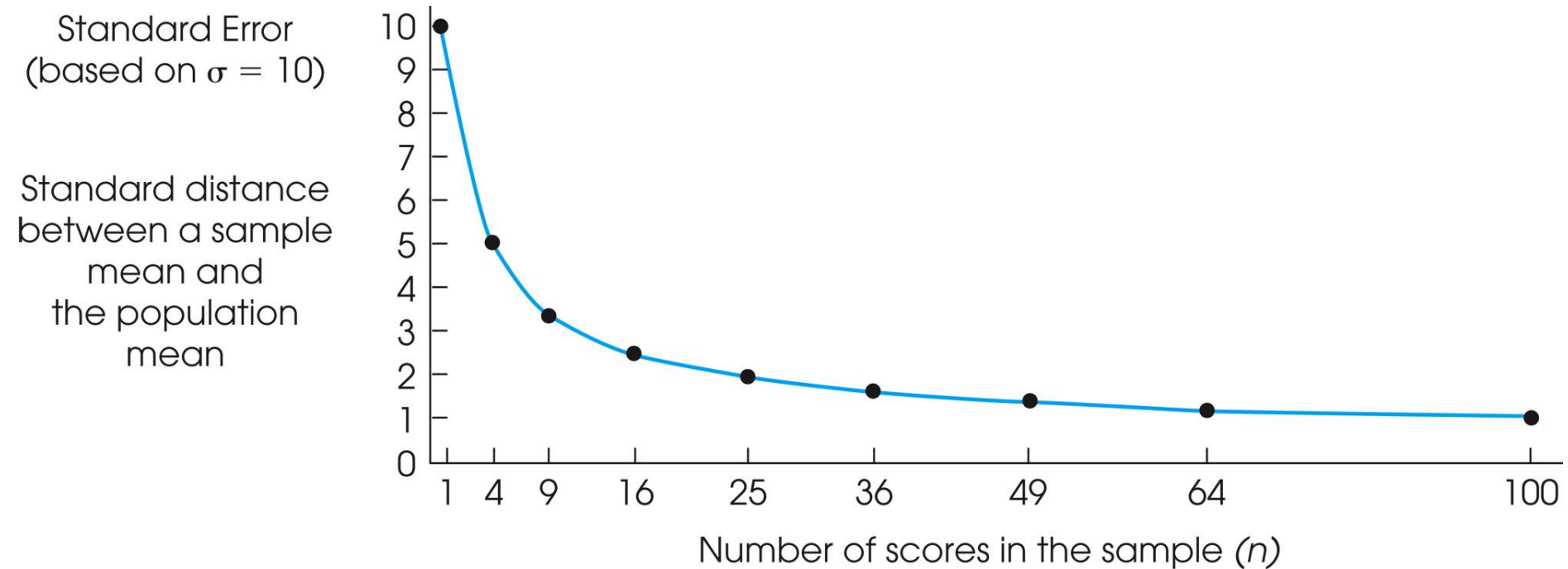
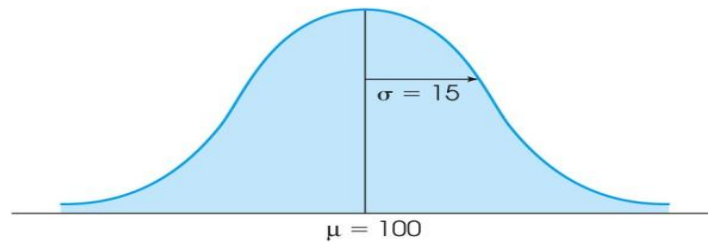
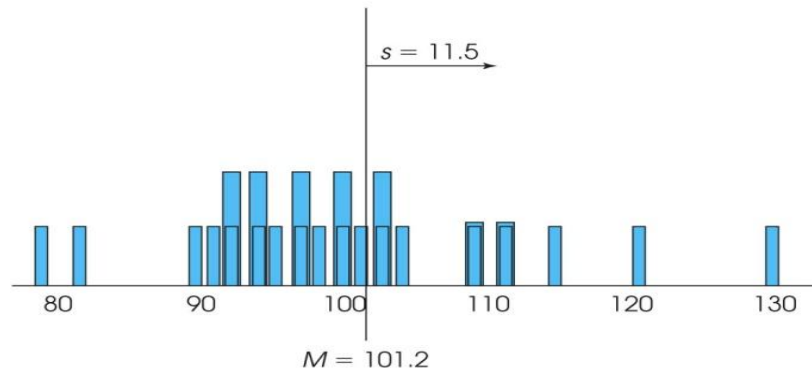


Figure 7.4 Distributions of Scores vs. Sample Means

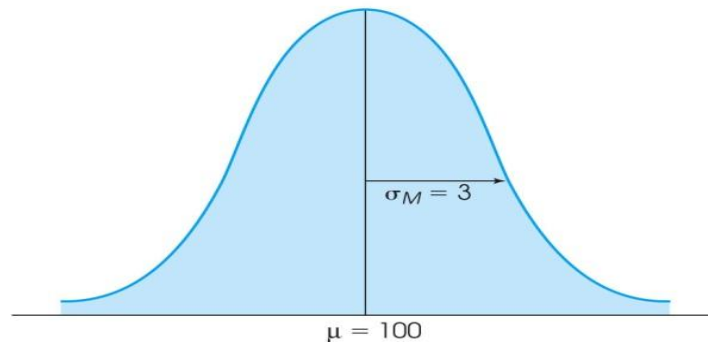
(a) Original population of IQ scores.



(b) A sample of $n = 25$ IQ scores.



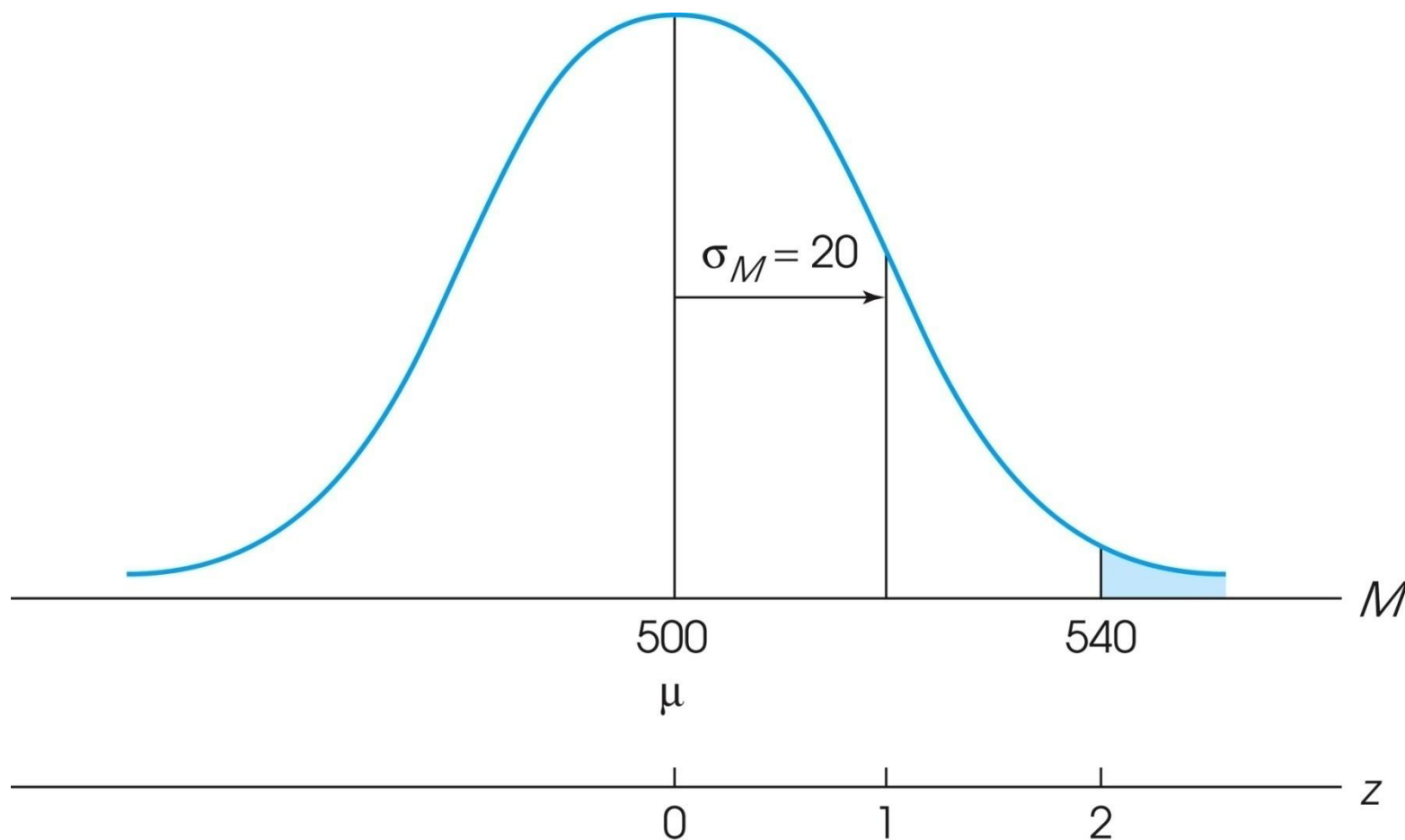
(c) The distribution of sample means. Sample means for all the possible random samples of $n = 25$ IQ scores.



7.3 Probability and the Distribution of Sample Means

- Primary use of the distribution of sample means is to find the probability associated with any particular sample (sample mean)
- Proportions of the normal curve are used to represent probabilities
- A z -score for the sample mean is computed

Figure 7.5 Distribution of Sample Means for $n = 25$

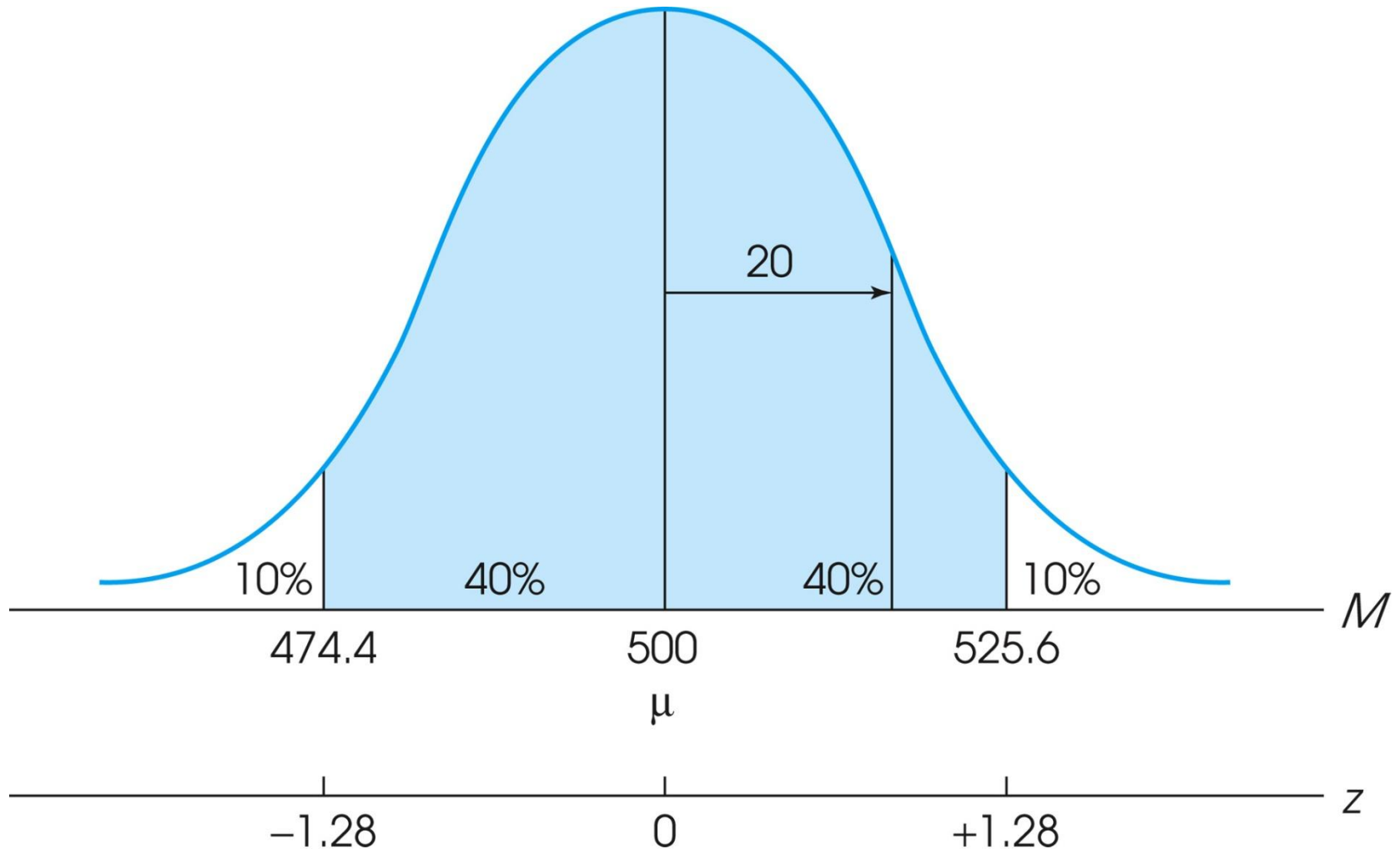


A z-Score for Sample Means

- Sign tells whether the location is above (+) or below (-) the mean
- Number tells the distance between the location and the mean in standard deviation (standard error) units
- z-formula:

$$z = \frac{M - \mu}{\sigma_M}$$

Figure 7.6 Middle 80% of the Distribution of Sample Means



Learning Check

- A population has $\mu = 60$ with $\sigma = 5$; the distribution of sample means for samples of size $n = 4$ selected from this population would have an expected value of _____

A

• 5

B

• 60

C

• 30

D

• 15

Learning Check - Answer

- A population has $\mu = 60$ with $\sigma = 5$; the distribution of sample means for samples of size $n = 4$ selected from this population would have an expected value of _____

A

• 5

B

• 60

C

• 30

D

• 15

Learning Check

- Decide if each of the following statements is True or False

T/F

- The shape of a distribution of sample means is always normal

T/F

- As sample size increases, the value of the standard error decreases

Learning Check - Answer

False

- The shape is normal *only* if the population is normal or $n \geq 30$

True

- Sample size is in the denominator of the equation so as n grows larger, standard error decreases

7.4 More about Standard Error

- There will usually be discrepancy between a sample mean and the true population mean
- This discrepancy is called sampling error
- The amount of sampling error varies across samples
- The variability of sampling error is measured by the standard error of the mean

Figure 7.7 Example of typical distribution of sample means

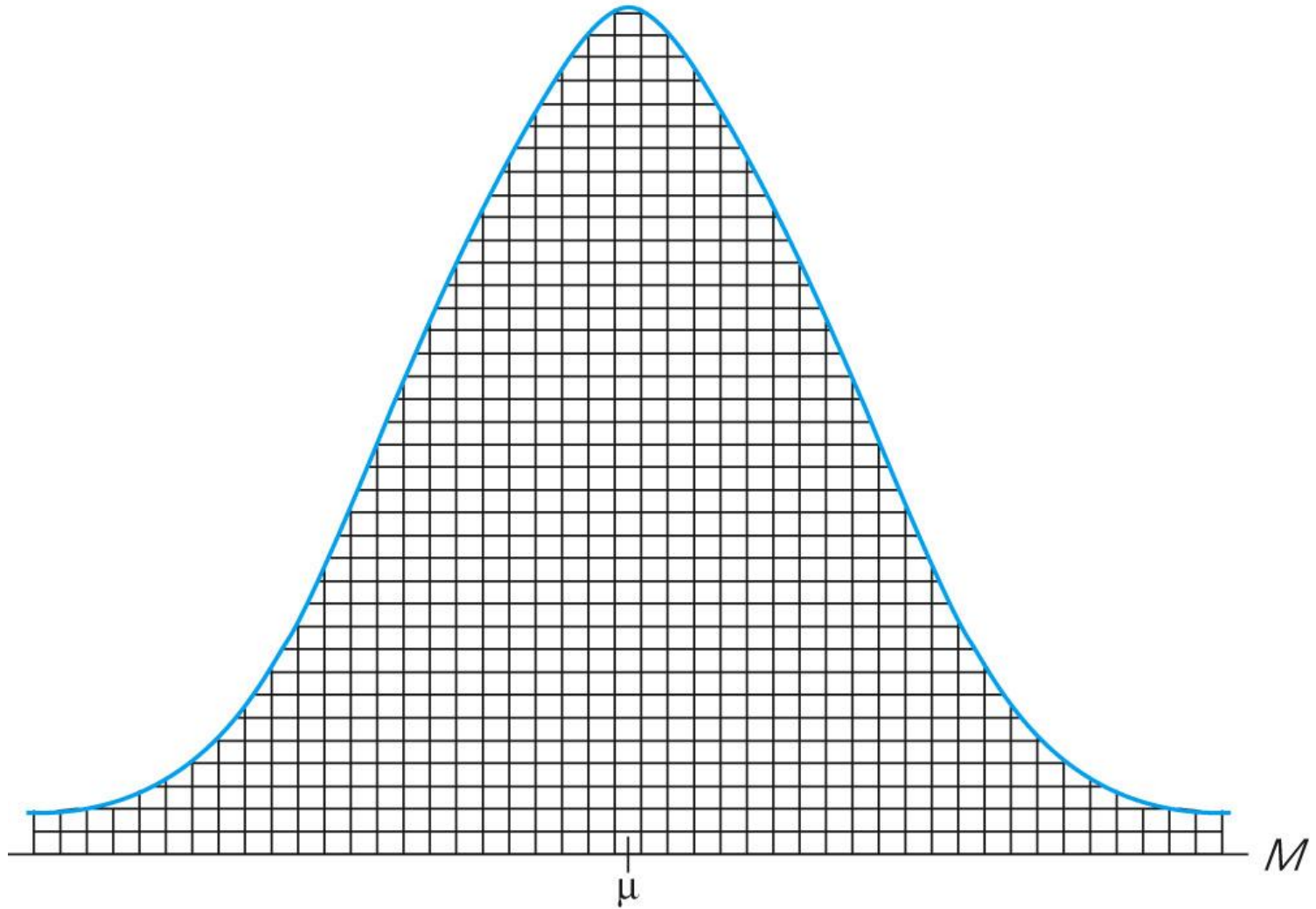
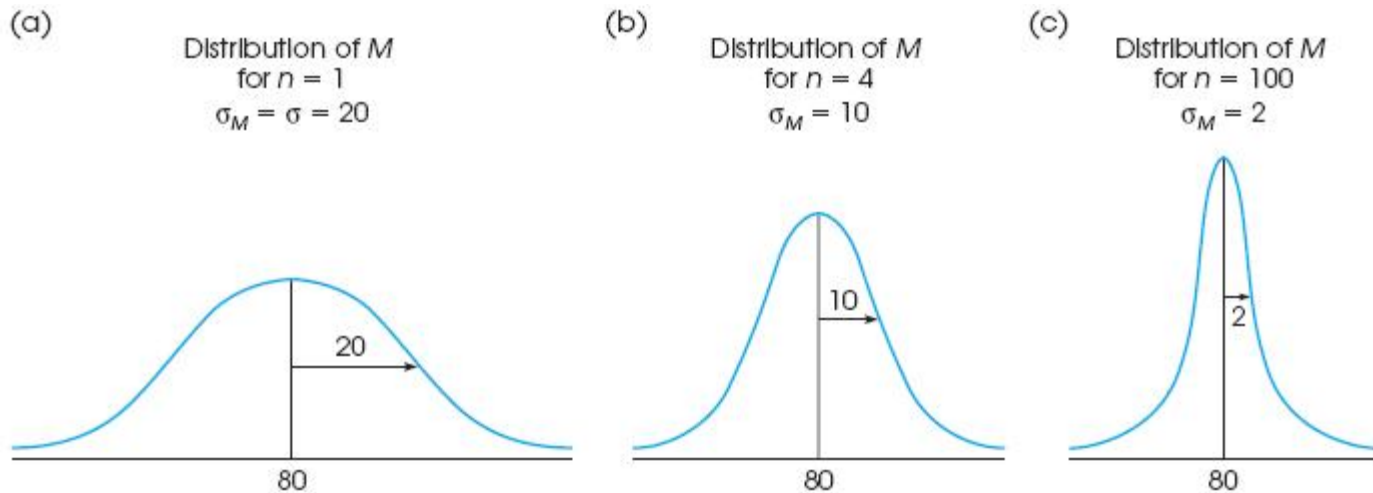


Figure 7.8 Distribution of Sample Means when $n = 1, 4,$ and 100



In the Literature

- Journals vary in how they refer to the standard error but frequently use:
 - SE
 - SEM
- Often reported in a table along with n and M for the different groups in the experiment
- May also be added to a graph

Figure 7.9
Mean ($\pm 1 SE$) in Bar Chart

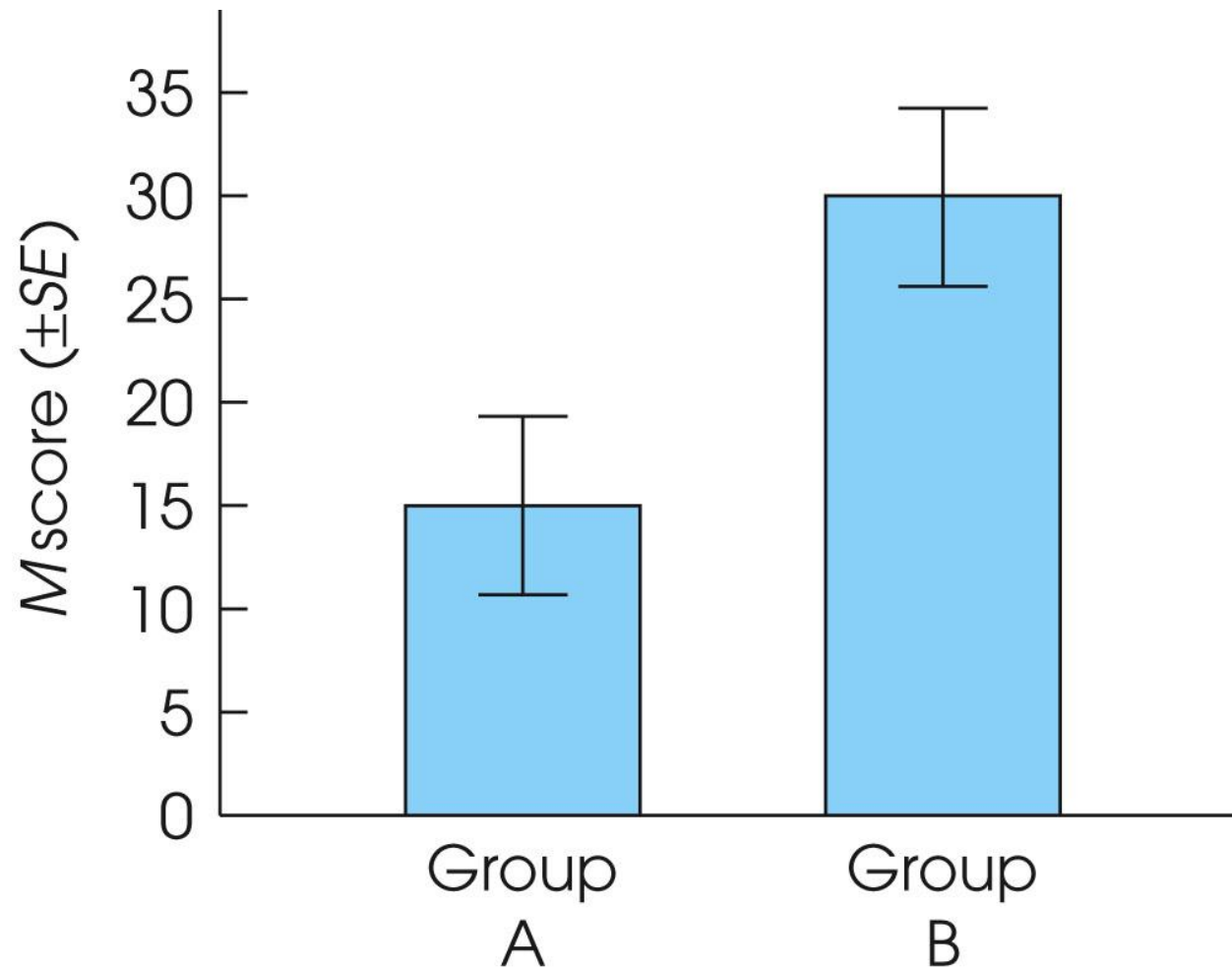
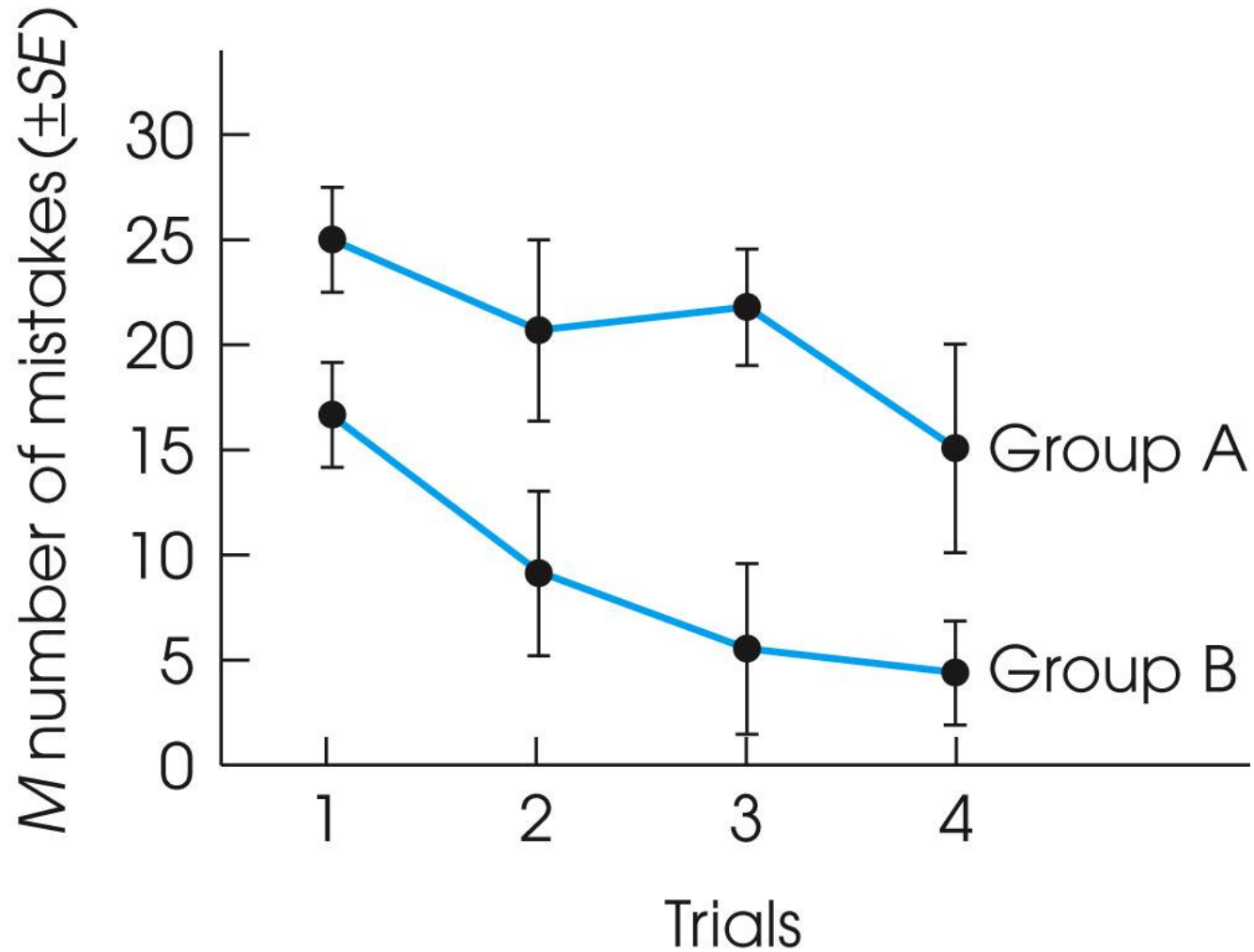


Figure 7.10
Mean ($\pm 1 SE$) in Line Graph



7.5 Looking Ahead to Inferential Statistics

- Inferential statistics use sample data to draw general conclusions about populations
 - Sample information is not a perfectly accurate reflection of its population (sampling error)
 - Differences between sample and population introduce uncertainty into inferential processes
- Statistical techniques use probabilities to draw inferences from sample data

Figure 7.11 Conceptualization of research study in Example 7.5

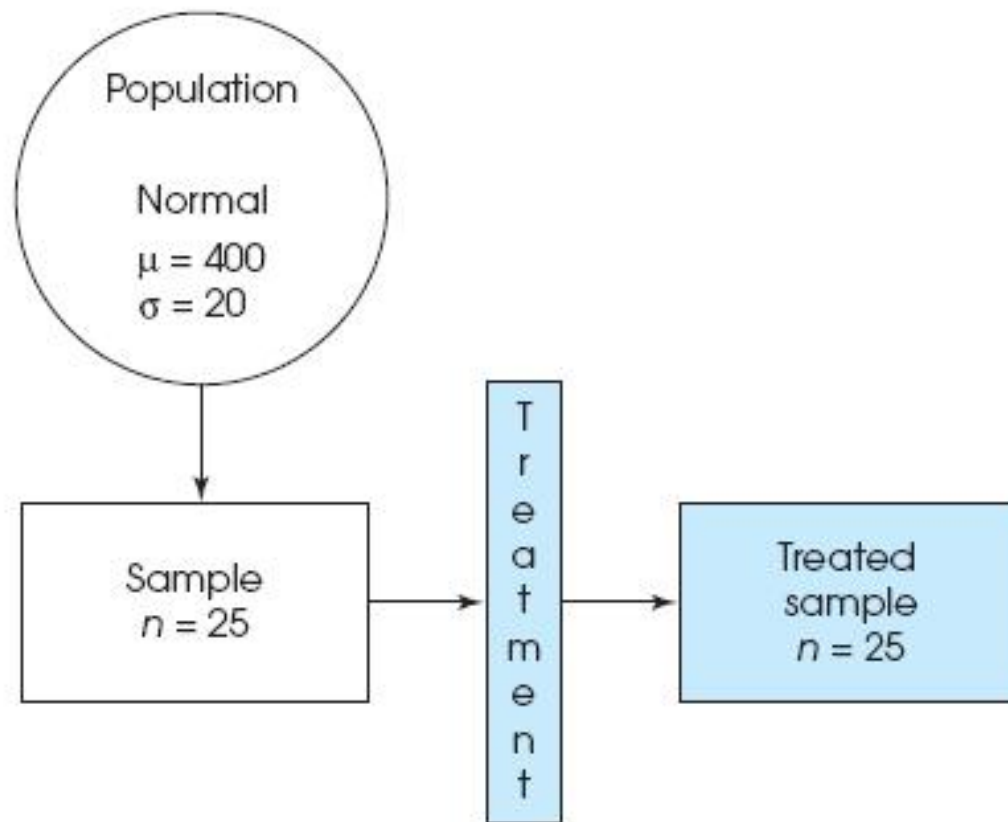
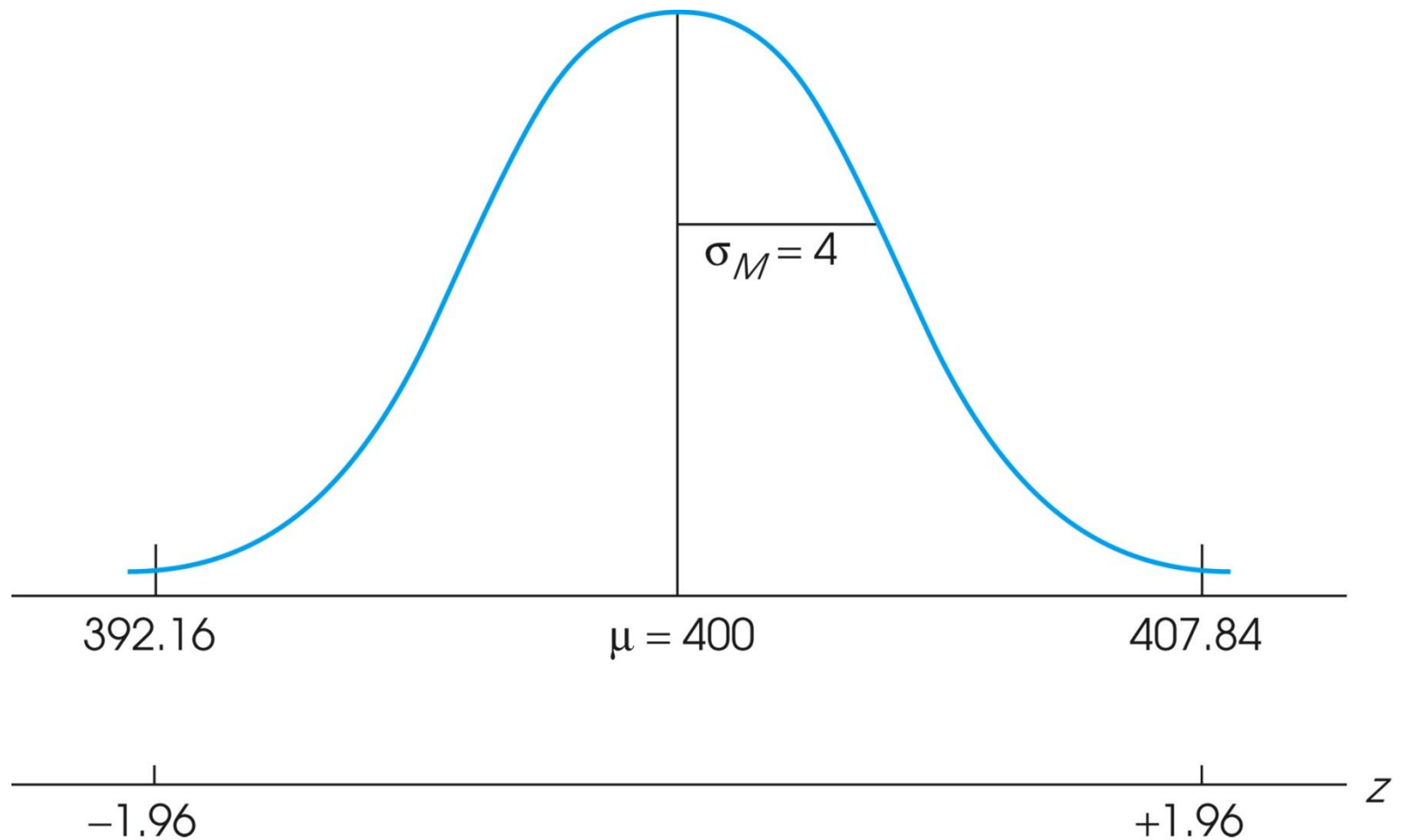


Figure 7.12 Untreated Sample Means from Example 7.5



Learning Check

- A random sample of $n = 16$ scores is obtained from a population with $\mu = 50$ and $\sigma = 16$. If the sample mean is $M = 58$, the z-score corresponding to the sample mean is ____?

A

- $z = 1.00$

B

- $z = 2.00$

C

- $z = 4.00$

D

- Cannot determine

Learning Check - Answer

- A random sample of $n = 16$ scores is obtained from a population with $\mu = 50$ and $\sigma = 16$. If the sample mean is $M = 58$, the z-score corresponding to the sample mean is _____?

A

• $z = 1.00$

B

• $z = 2.00$

C

• $z = 4.00$

D

• Cannot determine

Learning Check

- Decide if each of the following statements is True or False

T/F

- A sample mean with $z = 3.00$ is a fairly typical, representative sample

T/F

- The mean of the sample is always equal to the population mean

Learning Check - Answers

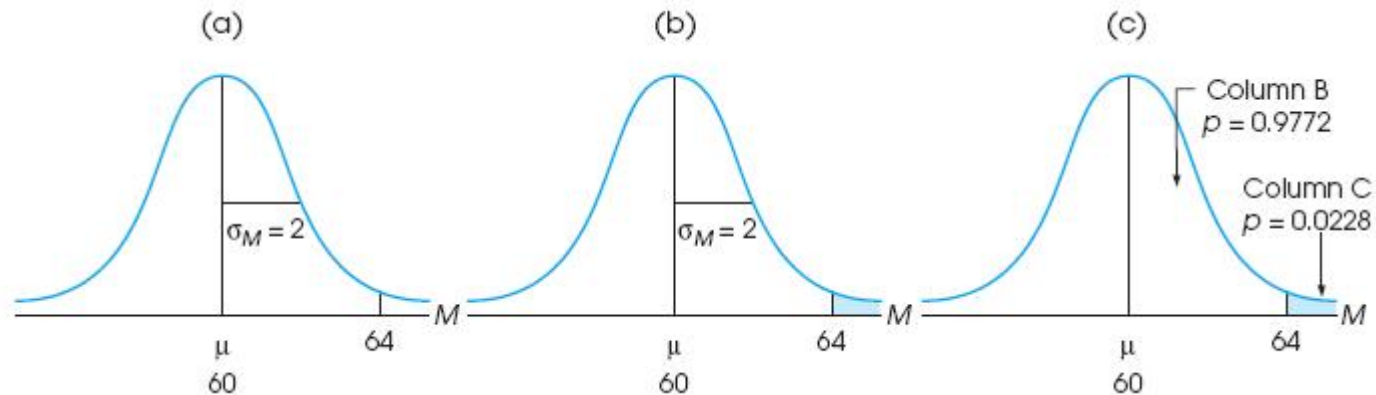
False

- A z-score of 3.00 is an extreme, or unlikely, z-score

False

- Individual samples will vary from the population mean

Figure 7.13 Sketches of Distribution in Demonstration 7.1



Homework

- Chapter 6
 - Q 2
 - Q 8
 - Q 14
 - Q20
- Chapter 7
 - Q 6
 - Q 12
 - Q 16
 - Q20